Paleomagnetism and the age of copper mineralization in the Portage Lake Volcanics, Upper Peninsula, Michigan

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The close association of hematite with native copper in the flow tops of the Portage Lake Volcanics invites the opportunity to establish the relative age of copper mineralization. For this purpose, we sampled 12 paleomagnetic sites representing six mineralized flow tops and their associated massive interiors on the Keweenaw Peninsula in Michigan. Three of the four less-altered flow interiors contained two components of magnetization: one was carried by primary magnetite, and therefore the other, carried by hematite, must be secondary. The hematite component was parallel to the single hematite component in the associated flow top, indicating that both components had a common secondary origin. The low between-site dispersion of the hematite component and the similarity of the mean direction to primary directions of other rocks of the same age (1100 Ma) suggest that hematization occurred over a period of time long enough to average out secular variation but before deposition of the overlying Nonesuch Shale, extensive tectonic tilting, and significant apparent polar wander. If copper mineralization and hematization were contemporaneous, then we have also dated the relative age of copper mineralization and the associated metamorphism. In contrast, paleomagnetic results from the two extensively altered flows (from below the Greenstone flow) are consistent with the mechanism of deuteric or auto-oxidation.

L'association étroite de l'hématite avec le cuivre natif aux sommets des coulées des volcanites du Portage Lake incite à rechercher un âge relatif de la minéralisation du cuivre. Pour ce faire, nous avons échantillonné 12 sites paléomagnétiques représentant six sommets de coulées minéralisées, incluant leurs intérieurs massifs associés, dans la région de la péninsule de Keweenaw au Michigan. Deux composantes d'aimantation caractérisent trois sur quatre des intérieurs des coulées les plus fraîches : une composante était portée par une magnétite primaire, et par conséquent l'autre, portée par de l'hématite, devrait être secondaire. La composante de l'hématite était parallèle à la composante unique trouvée dans l'hématite des sommets des coulées associées, ce qui indique pour les deux composantes une origine secondaire commune. Les valeurs faibles qui séparent les sites de dispersion de la composante de l'hématite et la convergence de la direction moyenne avec la directions primaires des autres roches de même âge (1100 Ma) indiquent que la formation de l'hématite a nécessité un temps suffisamment long qui a permis l'effacement des variations séculaires, mais elle est antérieure à l'accumulation des shales de Nonesuch susjacents, à l'inclinaison tectonique généralisée et à une dérive apparente des pôles significative. Si les événements de la minéralisation en cuivre et de l'hématisation étaient contemporains, alors l'âge relatif de la minéralisation en cuivre et du métamorphisme associé est donc établi. D'autre part, les résultats paléomagnétiques des deux coulées fortement altérées (sous-jacentes à la coulée Greenstone) temoignent plutôt d'arrets deutériques ou d'auto-oxydation.

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Introduction

Native copper deposits hosted by the Portage Lake Volcanics (PLV) of northern Michigan are clearly epigenetic (Butler and Burbank 1929; Stoiber and Davidson 1959; White 1968; Jolly 1974; Scofield 1976). The age of copper mineralization is estimated to be 1060 ± 20 Ma (Bornhorst *et al.* 1988) compared with the age of the PLV, which is between 1098 ± 4 Ma (age established by Davis and Sutcliffe (1985) of the Osler Group, thought to be older than the PLV) and 1054 ± 34 Ma (minimum age of the overlying Freda Sandstone, referred to in White (1968)). The imprecision of radiometric dating techniques leaves the timing of mineralization uncertain relative to deposition of overlying sediments and to tectonic tilting.

We may be able to determine the timing of copper mineralization using paleomagnetic methods if hematization occurred during mineralization. Native copper in the PLV is often found within highly altered, amygdaloidal, and brecciated flow tops in close association with hematite. Jolly (1974) and Scofield (1976) concluded that the alteration of the flow tops during copper mineralization resulted in oxidation of the primary magnetites to secondary hematites. The remanent direction associated with magnetite is thought to record the direction of the Earth's magnetic field during initial cooling of the lavas. If Jolly (1974) and Scofield (1976) are correct, however, during alteration and mineralization the original magnetite remanence in the flow tops was destroyed and replaced by a secondary hematite remanence. The purpose of our study is to determine the relative timing of copper mineralization by comparing the paleomagnetic directions of the mineralized flow tops with those of the less-altered interiors. To accomplish our goal, we first have to establish that the hematite was secondary and not produced during auto-oxidation as suggested by Butler and Burbank (1929).

General geology

The Portage Lake Volcanics are thought to be the product of an extensive rift system that existed 1000–1200 Ma ago (White 1960; Halls 1978; Green 1982). White (1960) suggested that these tholeiitic flood basalts erupted into the slowly subsiding rift basin from a fissure system located near the center of present-day Lake Superior and that at least 20° of total present-day dip is due to progressive tilting during deposition. As volcanism waned, sediments began to fill the basin with at least four major sedimentary units deposited on top of the PLV: the Copper Harbor Conglomerate, Nonesuch Shale, Freda Sandstone, and Jacobsville Sandstone (Fig. 1). Sometime after deposition of the Nonesuch Shale, the Keweenaw Fault, a low- to high-angle reverse fault, formed nearly parallel to PLV bedding and thrust PLV over the younger Jacobsville Sandstone (White 1968). During faulting, the lavas were tilted

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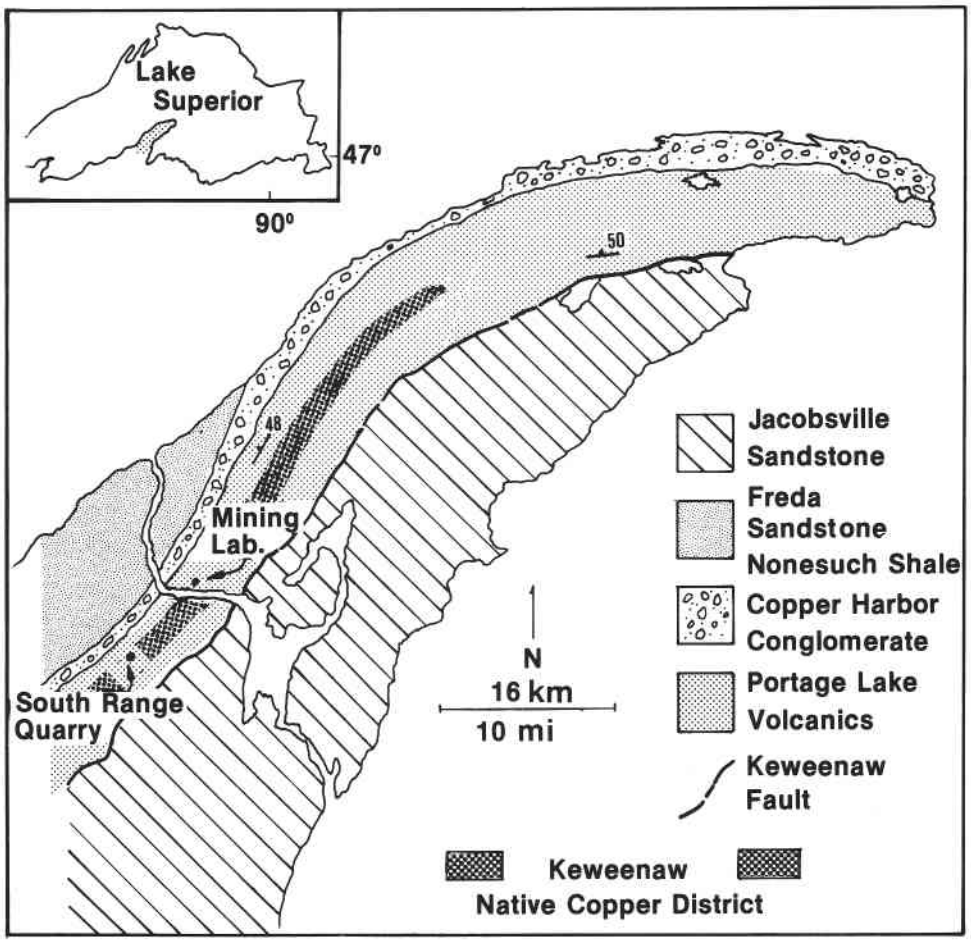


FIG. 1. Geologic map of the Keweenaw Peninsula, Michigan, showing the location of sampling sites within the Keweenawan native copper district. Geologic units are listed in stratigraphic order.

even more steeply towards the basin. Today, the entire section on the Keweenaw Peninsula dips to the northwest and north, forming the southern limb of the Lake Superior Syncline.

The volcanic and sedimentary pile was subjected to metamorphism and copper mineralization at some time during these events. At the base of the PLV, the once porous flow tops are metamorphosed as high as prehnite–pumpellyite facies, and the maximum metamorphic temperatures and pressures reached were 280°C and 1.5 kbar (1 kbar = 100 MPa) (Livnat 1983). The less-permeable flow interiors, especially the thicker ones, were largely unaffected by metamorphism and were relatively isolated from hydrothermal fluids. Copper also formed in large quantities in the Nonesuch Shale and in small amounts throughout the Copper Harbor Conglomerate and Freda Sandstone.

Sampling and laboratory procedures

Ninety-two samples representing six mineralized flow tops and their associated interiors were collected from the adit of the Michigan Technological University (MTU) mining laboratory (TM sites) in Hancock, Michigan, and from the South Range (SR site) quarry located near South Range, Michigan (Fig. 1). Stratigraphic locations of the sampling sites are given in Browning (1986). Strike and dip of beds are 29°E, 51°NW at the MTU mine and 60°E, 60°NW at South Range quarry. Flow thickness ranged from 11 m (sites TM7 and TM8) to 60 m (SR site). We carefully chose those flows whose tops were extremely altered and amygdaloidal, with interiors that exhibited a range of alteration.

In five of the flows, the tops were easy to locate. However, there were three amygdaloid-rich zones between flow interior TM9 and top TM11, which may represent tops of cooling units. Therefore, TM11 may not belong to the same cooling unit as flow interior TM9, a fact that may explain the paleomagnetic results discussed later.

All cores were drilled with a portable drill and oriented *in situ* with a Brunton compass attached to an orienting tool. Cores from the MTU mining adit were reoriented off a surveying line, whereas those from the quarry were checked with a sun compass.

The oriented cores were cut into 2.4 cm long cylinders. Their natural remanent magnetization (NRM) was measured using a Schonstedt SSM-1A spinner magnetometer. Then they were subjected to a detailed thermal demagnetization using a noninductively wound furnace controlled by an Omega 149 temperature controller. All cooling was done in three nested mu-metal cylinders with an ambient field of approximately 10 nT. Magnetic susceptibility measurements were made before and after each heating using a Barington meter. Alternating-field (AF) demagnetization on sister samples for sites TM12 and TM13 was completed with a Schonstedt GSD-1 AC demagnetizer.

Additional rock magnetic experiments on representative

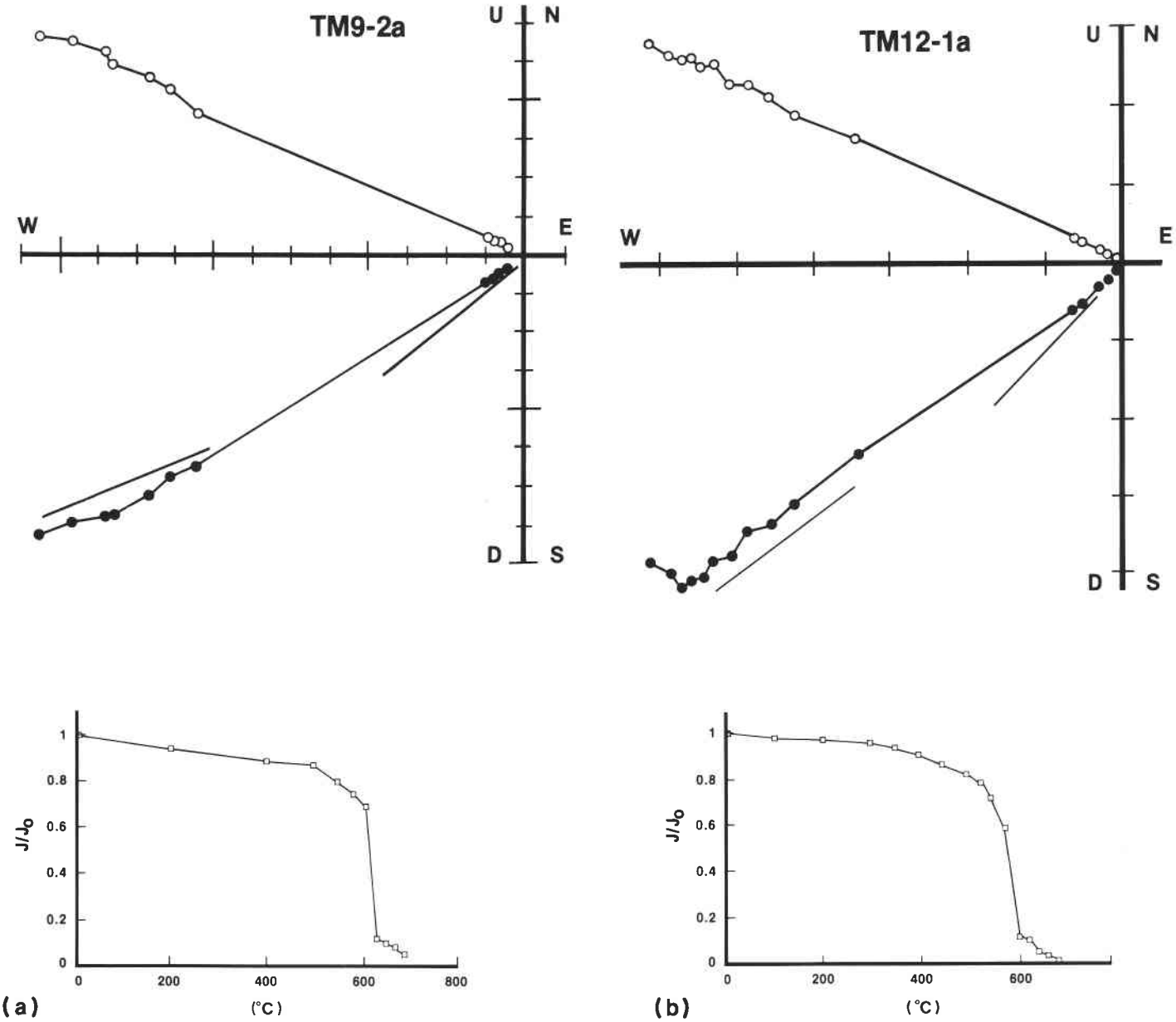


FIG. 2. Representative thermal demagnetization vector end-point diagrams and normalized intensity plots of group 1 flow interiors showing typical two-component behavior. The data in this and the following figures are tectonically rotated to the horizontal. Closed (open) circles represent the vertical (horizontal) projection. (*a*) Sample TM9-2a demagnetized at 0, 200, 400, 500, 550, 585, 610, 630, 650, 670, and 690°C. Large-scale divisions, 1 A/m (1×10^{-3} emu/cm^3). (*b*) Sample TM12-1a demagnetized at 0, 100, 195, 245, 340, 390, 440, 490, 515, 535, 575, 595, 615, 635, 655, and 675°C. Scale division, 0.174 A/m.

samples from each site were completed including isothermal remanent magnetization (IRM) acquisition studies and petrographic studies of polished thin sections. Hysteresis measurements and thermomagnetic analysis of flow interiors were made using a Princeton Applied Research vibrating sample magnetometer; thermomagnetic analysis of tops was completed using a Curie balance. Heating was done in air for highly altered samples to prevent reduction of hematite to magnetite (Shive and Diehl 1977).

Analytical studies of demagnetization behavior began with the visual inspection of vector end-point diagrams for each specimen (Zijderveld 1967). Best-fit directions to the components were calculated using the least-squares component analysis developed by Kirschvink (1980). For the flow interiors, one component was determined from around 200 to <585°C and a second for levels >585°C. Only one component was determined for the flow tops generally between 200 and 690°C, except for tops TM2 and TM4, for which higher temperature intervals were used. All of the directions passed the 10° maximum angular deviation (MAD) criterion of Kirschvink (1980). Finally, a Fisherian statistical analysis was completed (Fisher 1953).

Oxide mineralogy

Petrographic analysis of representative samples indicates that the Fe–Ti alteration phases can be attributed to two major events: (1) initial deuteric oxidation to class 2–4 of Ade-Hall

et al. (1971) and to stage C4-C5 and R3−R5 of Haggerty (1976) and (2) low-grade metamorphism. These results are similar to those of Ade-Hall *et al.* (1971) for the Keweenawan Mamainse Point lavas. Distinguishing between these two events will not be attempted here. A more detailed description is given in Browning (1986). The flow interiors were found to fall into three groups of oxidation, with group 1 being the least altered and group 3 being the most altered.

The increase in alteration from group 1 to 3 can be easily observed in the magnetite. Magnetite phenocrysts of the group 1 interiors contain abundant [111] exsolution lamellae. The interlamellar portions contain mottled magnetite, with parts oxidized to hematite. In group 2 interiors, abundant pseudomorphs after magnetite contain relic exsolution lamellae, with the intervening material almost completely oxidized to hematite. Group 3 flow interiors contain few, if any, remnants of primary magnetite. When present, magnetite is extremely altered to hematite and other oxide phases and appears as skeletal ghosts of the primary crystal. In all three groups; both the ilmenite exsolution lamellae and ilmenite phenocrysts are altered to intergrowths of hematite and rutile.

The flow interiors show progressively greater hematization staining of the matrix from group 1 to 3. White crystalline hematite alteration also becomes more extensive until in group 3 it covers the olivine ghosts and surrounds most of the altered oxides.

The flow tops are all highly altered to hematite, with no recognizable primary oxides and few iron silicates remaining. The tops associated with the less-altered group 1 and 2 flow interiors contain more crystalline hematite than those associated with group 3 interiors.

Paleomagnetic results

Thermal demagnetization

Behavior during thermal demagnetization generally depends upon the degree of alteration. Two components of magnetization are most prevalent in the least-altered, group 1 flow interiors; are harder to discern in group 2; and are only present in a few instances in the most-altered, group 3 interiors.

The components in group 1 interiors have thermally discrete blocking temperatures (Fig. 2): the first: a component that is removed between 200 and 585°C; the second, a steeper component with blocking temperatures between 610 and 690°C. Components in these temperature ranges suggest that the first component is carried by magnetite and the second by hematite. The majority of the remanence is lost around 585°C (Fig. 2), indicating that much of the NRM is carried by magnetite. Group 2 flow interiors also reveal two-component behaviour, but the magnitude of the low-temperature component is smaller than in group 1 interiors (Fig. 3*a*). Group 3 flow interiors rarely show more than one dominant direction (Fig. 3*b*), although flow interior TM3 does contain two samples with more complex demagnetization paths.

Flow tops associated with groups 1 and 2 interiors contain only one component of magnetization between 200 and 690°C (Fig. 4). Most of the remanence is lost between 670 and 690°C, indicating that the carrier is hematite. Flow tops associated with group 3 interiors contained two components of magnetization. The low-blocking-temperature component has declinations from 265 to 280° and inclinations from 10 to 25°, similar to directions for the overlying Nonesuch Shale and Freda Sandstone.

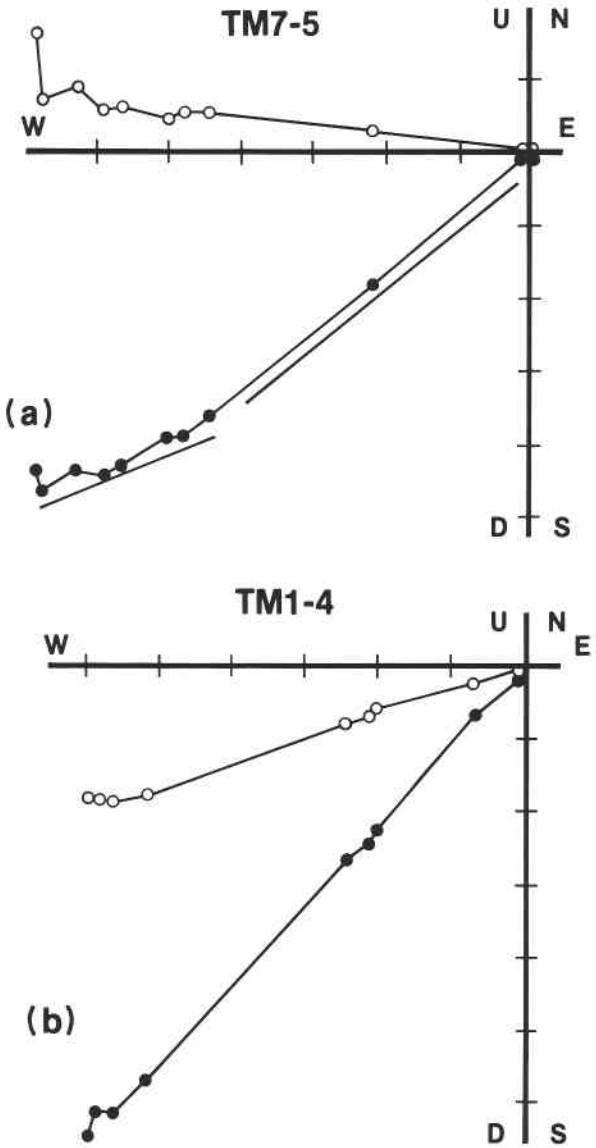


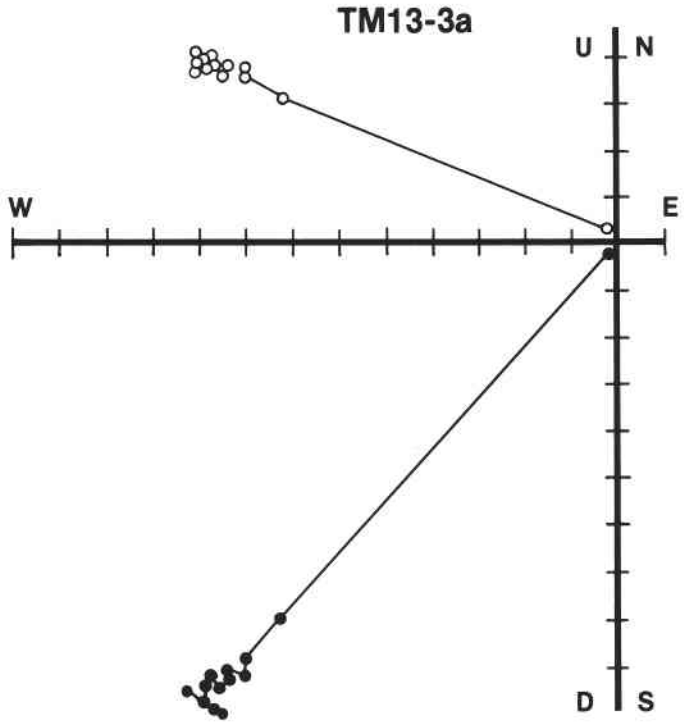
FIG. 3. Representative thermal demagnetization vector end-point diagrams for (*a*) group 2, sample TM7-5, demagnetized at 0, 215, 420, 500, 550, 585, 610, 630, 650, 670, and 690°C, with scale division of 2.4×10^{-2} A/m; and (*b*) group 3, sample TM1-4, demagnetized at 0, 200, 400, 500, 550, 585, 610, 630, 650, and 670°C, with scale division of 6.6×10^{-2} A/m. Symbols as in Fig. 2.

Susceptibility

Half the interiors and most flow tops showed an increase in susceptibility by as much as a factor of four after heating above 600°C because of the formation of new magnetic material. The samples were purposely oriented unsystematically while heating and cooling, although the long axes of the samples were always oriented parallel or antiparallel to the long axis of the mu-metal chambers. Under these circumstances, introduction of a spurious component of magnetization by the production of magnetic material should increase the dispersion of the directions. No such increase was observed, suggesting that laboratory-induced components were insignificant.

Alternating-field demagnetization

During AF demagnetization, the inclination of group 1 flow interior TM12 shows a curved decay path, indicating the pres-



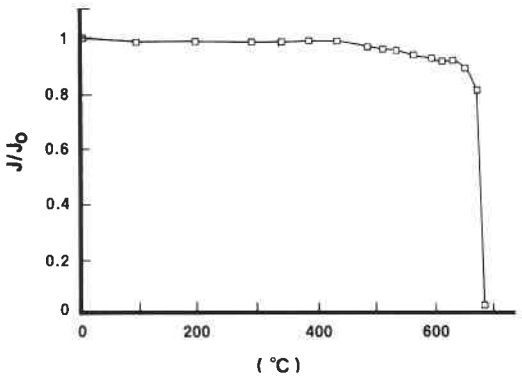


FIG. 4. Typical thermal demagnetization vector end-point diagram and normalized intensity plot for a mineralized flow top. Sample TM13-3a demagnetized at 0, 130, 295, 345, 390, 440, 490, 525, 535, 565, 595, 605, 635, 655, 675, and 685°C. Scale division, 0.136 A/m. Symbols as in Fig. 2.

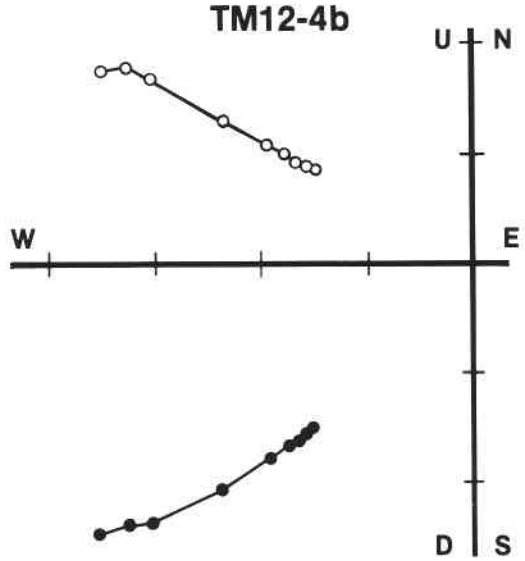


FIG. 5. Representative AF demagnetization vector end-point diagrams of flow interior TM12. Sample TM12-4b demagnetized at 0, 5, 10, 20, 30, 40, 50, 60, 70, 80, and 90 mT (× 10 Oe). Scale division, 0.36 A/m. Symbols as in Fig. 2.

ence of two components with overlapping coercivity spectra (Fig. 5). The mean inclination gradually increases with increasing fields. The mean inclinations are 34.2, 37.2, 41.3, and 42.2° at 10, 20, 40, and 90 mT, respectively. This same increase is seen during thermal demagnetization (Table 1). A large portion of the remanence (48–75%) remains after 90 mT. In the flow top, 88% of the remanence remained after demagnetization in fields of 90 mT.

Isothermal remanent magnetization

Results of IRM acquisition studies were consistent with thermal results. Group 1 flow interiors reached greater than 90% IRM saturation in magnetizing fields less than 150 mT, behaviour typical of magnetite (Dunlop 1972). Group 2 flow interiors and some group 3 interiors (TM1) behaved in an intermediate manner, reaching 50–70% IRM saturation in fields of 150 mT. Low-coercive magnetite is still dominant, but a high-coercive phase, hematite, is also present in significant quantities. Group 3 flow interiors only reached 40% IRM saturation in 150 mT, indicating that a high-coercive phase, most likely hematite, is the main magnetic mineral. The flow tops and some group 3 flow interiors did not reach saturation even in fields as great as 1000 mT, behavior typical of fine-grained hematite (Dunlop 1972).

Curie temperature and hysteresis

As with the preceding results, Curie temperature and hysteresis properties of representative samples were closely related to the amount of primary oxide alteration. The flow-interior thermomagnetic curves gave magnetite Curie temperatures of 585°C, with large increases in magnetization upon cooling in many of the samples. The single 585°C Curie temperature upon cooling suggests that the production of magnetite (probably from hematite) caused the large increase in magnetization and susceptibility during heating above 600°C. A change in slope around 350°C in one of the interiors during both heating and cooling suggests the presence of titanomagnetite. None of the thermomagnetic curves exhibit the "kink feature" at 200°C observed in the Mamainse Point lavas by Ade-Hall *et al.* (1971). Mineralized flow tops were found to have typical hematite Curie temperatures at approximately 690°C, but they also exhibited a minor inflection at 580°C, indicating the presence of a small amount of magnetite. Hysteresis parameters for all group 1 flow interiors suggested that the magnetite was pseudo-single domain (Day *et al.* 1977).

Paleomagnetic directions

Fisher statistics for both the low- and high-blocking-temperature components are shown in Table 1. Data from all samples are included except for two samples from site TM3, which have sample directions more than four standard deviations from the mean of the remaining samples and remanent

TABLE 1. Summary of paleomagnetic results for the Portage Lake Volcanics

Site	Demag. T (°C)	N	Direction D (°)	Direction I (°)	A (°)	K	Δ(°)	Pole Lat. (°N)	Pole Long. (°E)
TM1 int.	200–550	7	255.5	45.5	5.9	106	7.3	10.4	210.3
(Group 3)	585–690	7	250.3	47.5	2.8	460	3.5	8.6	215.0
TM2 top	585–690	7	250.2	49.7	2.4	642	3.0	10.0	216.3
TM3 int.	220–585	4	257.3	43.3	5.8	253	4.4	10.2	207.9
(Group 3)	630–690	4	251.4	44.9	9.7	90	7.4	7.5	212.8
TM4 top	610–690	6	252.7	39.0	3.1	466	3.4	5.0	209.0
TM7 int.	215–550	6	276.8	19.2	5.5	151	6.0	11.9	183.5
(Group 2)	585–690	6	283.1	38.2	3.4	401	3.7	24.3	187.7
TM8 top	215–670	7	282.2	35.9	3.2	363	3.9	22.5	186.1
TM9 int.	200–570	6	290.9	28.8	8.2	68	9.0	25.3	177.2
(Group 1)	615–675	6	289.3	35.7	4.4	230	4.9	27.4	181.8
TM11 top	200–690	6	279.9	45.6	3.0	518	3.2	26.0	194.4
TM12 int.	200–565	7	298.5	32.6	2.5	565	3.2	32.2	173.1
(Group 1)	595–675	7	295.7	40.6	2.9	440	3.6	34.1	180.0
TM13 top	130–690	8	296.1	40.9	1.8	895	2.5	34.5	180.0
SR int.	200–550	8	300.0	21.7	6.3	79	8.5	28.5	166.8
(Groups 1–2)	585–675	8	297.1	33.0	3.0	337	4.1	31.4	174.5
SR top	200–650	7	295.9	32.5	3.5	298	4.4	30.3	175.1
Mean	~200–<585	4	291.4	25.9	13.2	50	10.0	24.4	175.5
TM7,9,12, SR	>585–~690	4	291.3	37.0	6.9	180	5.2	29.3	181.1

NOTES: "Group" refers to the petrological classification discussed in text. Temperature intervals for which best-fit directions were calculated are listed under Demag. A, K, and Δ are the radius of the circle of 95% confidence, precision parameter, and angular standard deviation, respectively. Site locations are 47.0° N, 271.7° E for TM sites and 47.1° N, 271.4° E for the SR site.

TABLE 2. Summary of paleomagnetic poles for the sediments overlying the Portage Lake Volcanics

Unit	Component	Declination (°)	Inclination (°)	Latitude	Longitude	N	Reference
Copper Harbor	P	289.9	37.2	28.5	183.1	20	Haig *et al.* (1985)
	S	278.9	14.7	11.5	180.7	3	Palmer *et al.* (1981)
	P*	297.1	−7.6	15.3	157.5	6	Elmore and Van der Voo (1982)
	S	286.5	0	11.2	168.3	7	Elmore and Van der Voo (1982)
Nonesuch	P	279.8	9.8	10.3	176.5	(27)	Henry *et al.* (1977)
	S	280.6	−9.5	3.7	169.2	(34)	Henry *et al.* (1977)
Freda	P	271.3	0.7	1.2	179.5	(85)	Henry *et al.* (1977)
	S	280.6	−9.5	3.7	169.2	(34)	Henry *et al.* (1977)

NOTES: N with (without) parentheses is the number of samples (sites). P and S are the primary and secondary components.
*Elmore and Van der Voo (1982) suggested the pole position is questionable.

intensities an order of magnitude weaker. These factors indicate that the samples did not have the same magnetic history as the rest of the site; therefore, they were excluded from the site mean. The directions of the high-temperature components of these samples were $D = 266°$, 269° and $I = 12°$, 14°, very similar to the directions in the Nonesuch Shale and Freda Sandstone (Table 2).

In three of the four sites from the less-altered groups 1 and 2, the 95% circle of confidence of the low-blocking-temperature component does not intersect that of the high-temperature component; therefore, the directions are statistically different (northwest quadrant in Fig. 6). In the fourth site, TM9, the hematite mean falls at the edge of the magnetite circle of confidence and according to the F test (McFadden and Lowes 1981), the kappas of the two components are not equal at the 95% confidence level. We applied the test of McFadden and Lowes (1981) to determine if the means of the two populations were different when the relationship between the kappas was unknown, but the result of this test was not conclusive. In this situation, we cannot statistically determine whether the mean hematite component in TM9 is the same as the mean magnetite component (McFadden and Lowes 1981). According to the hypothesis we propose, there is reason to expect that the primary and secondary components of a few flows would be similar. Another possibility for the directional similarity is explored in the next paragraph.

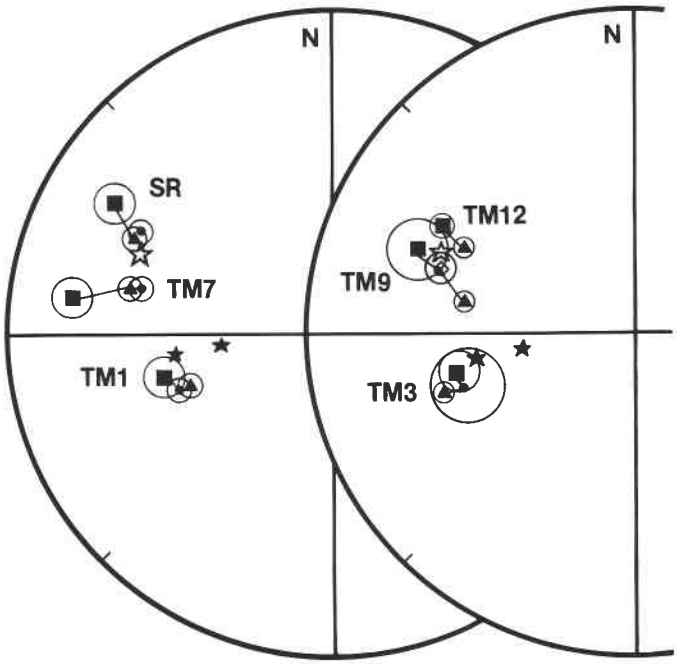


FIG. 6. Equal-area projection of site mean directions for the magnetite (squares) and hematite (circles) components of the flow interiors, the flow-top direction (triangles), and the 95% circles of confidence from Table 1. Based on the degree of oxidation, sites TM9, TM12, and part of SR belong to group 1; TM7 and part of SR, to group 2; TM1 and TM3, to group 3. Anomalous directions (closed stars) of Books (1972), mean PLV direction (open diamond) of Halls and Pesonen (1982), and mean Mamainse Point directions (open star) of Robertson (1973) are added for comparison. The mean hematite direction from the flow interiors falls on the Mamainse Point mean.

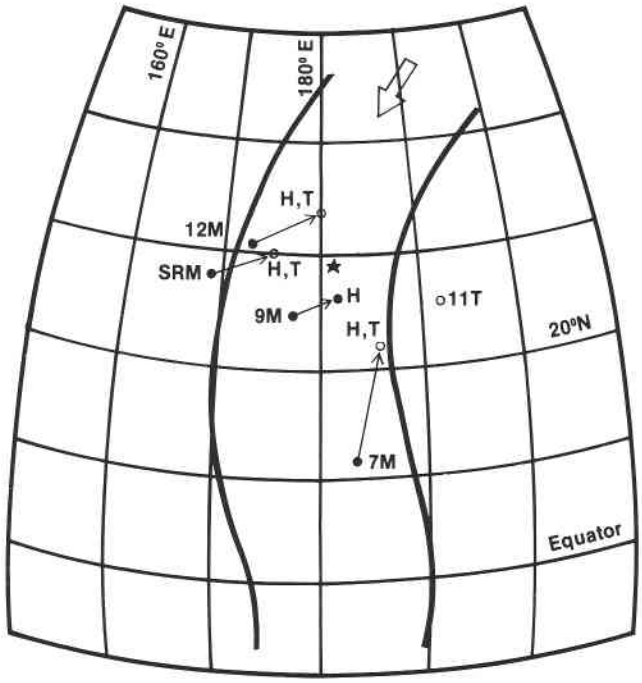


FIG. 7. Lake Keweenawan polar wander path for North America (after Morris and Roy 1977) with mean site pole positions for the magnetite (M) and hematite (H) components of group 1 and 2 flow interiors and for the associated flow top (T, open circles). Numbers represent flow-interior sites. The star represents the mean hematite component from the flow interiors. See text for a discussion about the relationship between sites TM9 and TM11.

The direction in three of the four flow tops is within 3° of the hematite direction from the associated interior. The exception is again TM9, where the hematite direction is statistically different from that of top TM11 (Fig. 6). As explained earlier, top TM11 may not be the same cooling unit as interior TM9. At site TM9, the directions at 610°C were intermediate between the magnetite and hematite directions. Thus, another explanation may be that the hematite remanence in the flow interior is a combination of both deuteric oxidation (primary) and alteration after the lava cooled.

Petrographic evidence confirms that the magnetite in the flow interiors is the result of deuteric oxidation and is primary. The basalt has not been heated above 300°C; therefore, the magnetization of the magnetite is also primary (using the formulae of Pullaiah *et al.* 1975). Because the magnetite and hematite components are not parallel, the hematite direction must be secondary. In addition, the similarity between the flow-top direction and the hematite direction in the associated interior strongly implies that they were acquired at the same time and that the directions in group 1 and 2 flow tops are also secondary.

In contrast, the low-blocking-temperature component in group 3 flow interiors falls very near both the hematite component in the interior and the flow-top direction (TM1 and TM3 in Fig. 6). Because IRM acquisition studies of TM1 indicate at least 50% of the remanence may be magnetite, magnetite is probably the carrier of the low-blocking-temperature remanence; thus, the hematite component in group 3 flows may be primary.

All paleomagnetic pole positions for group 1 and 2 flow interiors and flow tops were similar to other thermally demagnetized rocks of Keweenawan age outside the native copper district, e.g., the Mamainse Point lavas: longitude, 182.6°E; latitude, 29.4°N (Robertson 1973, as calculated in Halls and Pesonen 1982). In contrast, paleomagnetic directions for group 3 flow interiors and tops do not correspond to any "normal" Keweenawan direction. They are, however, very similar to anomalous flow directions discovered by Books (1972) for several other flows just below the Greenstone flow (Fig. 6). Because of their anomalous directions, group 3 flow interiors and their associated tops were not included in the determination of the mean magnetite and hematite directions (Table 1). Because only four magnetite directions were used in calculating a mean pole, this pole should not be regarded as a paleomagnetic pole.

Discussion and conclusions

The least-altered, group 1 and 2 flow interiors of the PLV contain two components of magnetization: (1) a primary magnetite component that is generally isolated between 200 and 585°C and (2) a more steeply inclined secondary hematite component between 585 and 690°C. The flow tops associated with these interiors were found to contain one dominant component of magnetization, which in three of the four flows was identical to the hematite component in the flow interior and was statistically distinct from the magnetite component. In these less-altered flows, therefore, most of the remanence-producing hematite could not be derived from deuteric or autooxidation, the mechanism suggested by Butler and Burbank

(1929) for the formation of the flow-top hematite throughout the PLV.

The possibility that one or the other of the components was introduced by the thermal demagnetization process can be eliminated because the cores were oriented as randomly as possible during cooling. In addition, AF demagnetization revealed the same components.

The timing of the acquisition of the secondary component can be deduced by the relative position of the magnetite and hematite components. If hematization took place during the Paleozoic or Late Precambrian when the basalts were tilted to their present attitude, the secondary hematite component would be more shallow than the primary magnetite component. As this is not the case, hematization must have occurred before tectonic tilting to present attitudes. If the basalts were tilted to the southwest (away from the center of the rift) and opposite to their present-day tilt direction prior to hematization, the hematite component should be more steeply inclined, as is found in our study. We cannot eliminate this possibility completely, but geologic evidence strongly favors continuous, moderate tilting toward the present-day attitude during deposition of the lavas and not in an opposite sense (White 1968). The steeper hematite components are probably not the result of apparent polar wander between basalt deposition and hematization, because the general movement between the primary and secondary components is perpendicular or opposite to the movement along the Keweenawan polar wander path (Fig. 7).

The most likely explanation is that hematization occurred before any significant tilting or apparent polar wander. The primary magnetite component was acquired over a relatively short period of time when the lavas cooled and records a point on the secular variation curve. The secondary hematite component would have been acquired over a time span sufficiently long to average out a large portion of the secular variation. The relatively small amount of scatter of the hematite directions ($K = 180$) compared with that of the magnetite directions ($K = 50$) is consistent with this hypothesis (Table 1). The consistently shallower magnetite directions may be due to chance; we just happened to sample flows that cooled when the Earth's secular variation was on the far side of the time-averaged pole for the Keweenawan. More flows need to be sampled to substantiate this conclusion.

The close similarity of the mean hematite pole position of group 1 and 2 flows to that of the Mamainse Point lavas (Robertson 1973) and that of the lavas in the overlying Copper Harbor Conglomerate (Haig *et al.* 1985) (Table 2) is also consistent with this conclusion. More specifically, the position of the pole suggests that the hematite component was acquired within either PLV or Copper Harbor time or both, before significant tectonic tilting.

Group 3 flow interiors revealed a different alteration history, with only one component of magnetization between 200 and 690°C giving a direction nearly identical to the high-temperature component in the top. The absence of two components and the presence of magnetite in flow TM1, as implied by IRM-acquisition curves, indicate that the hematite in these highly altered flows is primary and thus originated from deuteric oxidation. The anomalous direction cannot be explained by any simple combination of secondary components; therefore, we will assume it records an excursion of the Earth's magnetic field during PLV time.

Most earlier paleomagnetic studies on the PLV did not detect two components of magnetization. Those studies, in which

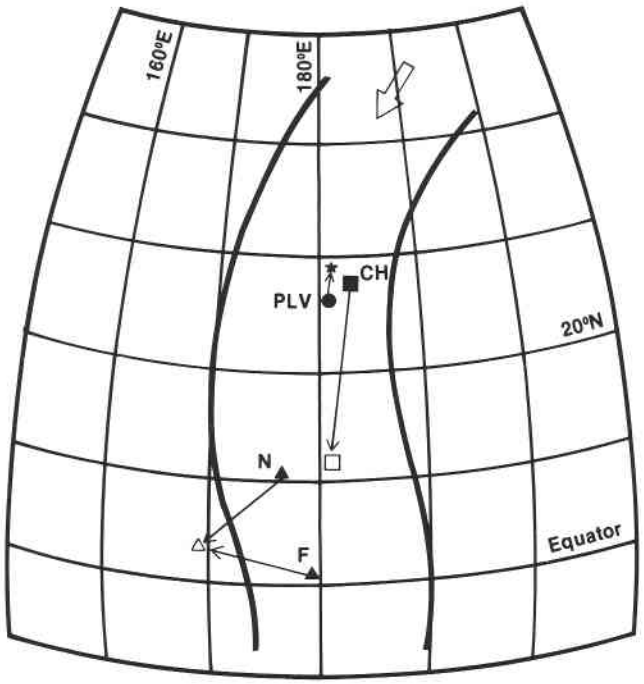


FIG. 8. Mean pole positions of primary and secondary components for the Keweenawan Copper Harbor (CH), Nonesuch (N), and Freda Sandstone (F). Closed (open) symbols of these three units represent poles interpreted as primary (secondary). For comparison, the PLV pole position from Books (1972) and the hematite component (star) from this study have been added.

demagnetization techniques were utilized, used pilot samples and blanket AF demagnetization to 10 mT (Books 1968, 1972) or thermal demagnetization that introduced spurious components (Vincenz 1968). Our results indicate that the remanent direction after 10 mT would be intermediate between the primary and secondary directions. If our hypothesis on the timing of hematization is correct, the PLV mean pole calculated by Halls and Pesonen (1982) using the data from Books (1972) would still be reliable, but the scatter would not accurately reflect paleosecular variation.

Dating copper mineralization in the PLV depends on the relationship between oxidation of the lava flows and copper mineralization. Some workers do not believe that these two events occurred together, e.g., Butler and Burbank (1929). However, Scofield (1976) found enough copper in the magnetite from the unaltered flow interiors to account for the native copper in the flow tops and hypothesized that oxidation of the magnetite released the copper. If Scofield (1976) and Jolly (1974) are correct and copper mineralization occurred during hematization, our hematite components indicate the time of copper mineralization, i.e., prior to significant tectonic tilting, shortly after extrusion of the lavas and before deposition of the overlying Nonesuch. The timing suggests that the metamorphism and mineralization were an integral part of the rifting process, similar to, but not identical with, the hydrothermal processes occurring in Iceland today (Walker 1960).

Secondary components of chemical origin have been identified by a number of authors in the sediments and a few lava flows overlying the PLV (Henry *et al.* 1977; Palmer *et al.* 1981; Halls and Palmer 1981; Elmore and Van der Voo 1982). The secondary components of these rocks are shallower than the primary ones (Table 1), implying that all the secondary

components for the Keweenawan rocks were acquired at different times (Fig. 8). Our data suggest that secondary hematization within the PLV occurred soon after initial deposition of the basalts and may have continued through Copper Harbor time. The secondary component from Palmer *et al.* (1981) for the Copper Harbor Conglomerate was acquired during Nonesuch time, and Henry *et al.* (1977) found that the secondary components for the Nonesuch and Freda seem to have been acquired during the Late Precambrian. If the secondary components were due to the mineralization event, as these workers suggest, then the mineralization was a continuous process beginning during the deposition of the PLV and continuing through Freda time. The occasional presence of secondary components in the PLV with directions similar to the primary components in the Nonesuch Shale and Freda are consistent with this conclusion.

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